

|                |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                |                                                                                                                                                                                                                            |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                 |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <div>678</div>                                                                                                                                                                                                             | <div>678</div>                                                                                                                                                                                                                 | <div>678</div>                                                                                                                                                                                                             | <div>678</div>                                                                                                                                                                                                             | <div>678</div>                                                                                                                                                                                                                 | <div>678</div>                                                                                                                                                                                                             | <div>678</div>                                                                                                                                                                                                                   |                                                                                                                                                                                                                                 |
| <div>345</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>abandon<br/>ability<br/>able<br/>about<br/>above<br/>absent<br/>absorb<br/>abstract</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>absurd<br/>abuse<br/>access<br/>accident<br/>account<br/>accuse<br/>achieve<br/>acid</div>    | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>acoustic<br/>acquire<br/>across<br/>act<br/>action<br/>actor<br/>actress<br/>actual</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>adapt<br/>add<br/>addict<br/>address<br/>adjust<br/>admit<br/>adult<br/>advance</div>     | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>advice<br/>aerobic<br/>affair<br/>afford<br/>afraid<br/>again<br/>age<br/>agent</div>         | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>agree<br/>ahead<br/>aim<br/>air<br/>airport<br/>aisle<br/>alarm<br/>album</div>           | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>alcohol<br/>alert<br/>alien<br/>all<br/>alley<br/>allow<br/>almost<br/>alone</div>              | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>alpha<br/>already<br/>also<br/>alter<br/>always<br/>amateur<br/>amazing<br/>among</div>        |
| <div>346</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>amount<br/>amused<br/>analyst<br/>anchor<br/>ancient<br/>anger<br/>angle<br/>angry</div>  | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>animal<br/>ankle<br/>announce<br/>annual<br/>another<br/>answer<br/>antenna<br/>antique</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>anxiety<br/>any<br/>apart<br/>apology<br/>appear<br/>apple<br/>approve<br/>april</div>    | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>arch<br/>arctic<br/>area<br/>arena<br/>argue<br/>arm<br/>armed<br/>armor</div>            | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>army<br/>around<br/>arrange<br/>arrest<br/>arrive<br/>arrow<br/>art<br/>artefact</div>        | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>artist<br/>artwork<br/>ask<br/>aspect<br/>assault<br/>asset<br/>assist<br/>assume</div>   | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>asthma<br/>athlete<br/>atom<br/>attack<br/>attend<br/>attitude<br/>attract<br/>auction</div>    | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>audit<br/>august<br/>aunt<br/>author<br/>auto<br/>autumn<br/>average<br/>avocado</div>         |
| <div>345</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>avoid<br/>awake<br/>aware<br/>away<br/>awesome<br/>awful<br/>awkward<br/>axis</div>       | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>baby<br/>bachelor<br/>bacon<br/>badge<br/>bag<br/>balance<br/>balcony<br/>ball</div>          | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>bamboo<br/>banana<br/>banner<br/>bar<br/>barely<br/>bargain<br/>barrel<br/>base</div>     | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>basic<br/>basket<br/>battle<br/>beach<br/>bean<br/>beauty<br/>because<br/>become</div>    | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>beef<br/>before<br/>begin<br/>behave<br/>behind<br/>believe<br/>below<br/>belt</div>          | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>bench<br/>benefit<br/>best<br/>betray<br/>better<br/>between<br/>beyond<br/>bicycle</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>bid<br/>bike<br/>bind<br/>biology<br/>bird<br/>birth<br/>bitter<br/>black</div>                 | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>blade<br/>blame<br/>blanket<br/>blast<br/>bleak<br/>bless<br/>blind<br/>blood</div>            |
| <div>346</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>blossom<br/>blouse<br/>blue<br/>blur<br/>blush<br/>board<br/>boat<br/>body</div>          | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>boil<br/>bomb<br/>bone<br/>bonus<br/>book<br/>boost<br/>border<br/>boring</div>               | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>borrow<br/>boss<br/>bottom<br/>bounce<br/>box<br/>boy<br/>bracket<br/>brain</div>         | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>brand<br/>brass<br/>brave<br/>bread<br/>breeze<br/>brick<br/>bridge<br/>brief</div>       | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>bright<br/>bring<br/>brisk<br/>broccoli<br/>broken<br/>bronze<br/>broom<br/>brother</div>     | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>brown<br/>brush<br/>bubble<br/>buddy<br/>budget<br/>buffalo<br/>build<br/>bulb</div>      | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>bulk<br/>bullet<br/>bundle<br/>bunker<br/>burden<br/>burger<br/>burst<br/>bus</div>             | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>business<br/>busy<br/>butter<br/>buyer<br/>buzz<br/>cabbage<br/>cabin<br/>cable</div>          |
| <div>345</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>cactus<br/>cage<br/>cake<br/>call<br/>calm<br/>camera<br/>camp<br/>can</div>              | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>canal<br/>cancel<br/>candy<br/>cannon<br/>canoe<br/>canvas<br/>canyon<br/>capable</div>       | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>capital<br/>captain<br/>car<br/>carbon<br/>card<br/>cargo<br/>carpet<br/>carry</div>      | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>cart<br/>case<br/>cash<br/>casino<br/>castle<br/>casual<br/>cat<br/>catalog</div>         | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>catch<br/>category<br/>cattle<br/>caught<br/>cause<br/>caution<br/>cave<br/>ceiling</div>     | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>celery<br/>cement<br/>census<br/>century<br/>cereal<br/>certain<br/>chair<br/>chalk</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>champion<br/>change<br/>chaos<br/>chapter<br/>charge<br/>chase<br/>chat<br/>cheap</div>         | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>check<br/>cheese<br/>chef<br/>cherry<br/>chest<br/>chicken<br/>chief<br/>child</div>           |
| <div>346</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>chimney<br/>choice<br/>choose<br/>chronic<br/>chuckle<br/>chunk<br/>churn<br/>cigar</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>cinnamon<br/>circle<br/>citizen<br/>city<br/>civil<br/>claim<br/>clap<br/>clarify</div>       | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>claw<br/>clay<br/>clean<br/>clerk<br/>clever<br/>click<br/>client<br/>cliff</div>         | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>climb<br/>clinic<br/>clip<br/>clock<br/>clog<br/>close<br/>cloth<br/>cloud</div>          | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>clown<br/>club<br/>clump<br/>cluster<br/>clutch<br/>coach<br/>coast<br/>coconut</div>         | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>code<br/>coffee<br/>coil<br/>coin<br/>collect<br/>color<br/>column<br/>combine</div>      | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>come<br/>comfort<br/>comic<br/>common<br/>company<br/>concert<br/>conduct<br/>confirm</div>     | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>congress<br/>connect<br/>consider<br/>control<br/>convince<br/>cook<br/>cool<br/>copper</div>  |
| <div>345</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>copy<br/>coral<br/>core<br/>corn<br/>correct<br/>cost<br/>cotton<br/>couch</div>          | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>country<br/>couple<br/>course<br/>cousin<br/>cover<br/>coyote<br/>crack<br/>cradle</div>      | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>craft<br/>cram<br/>crane<br/>crash<br/>crater<br/>crawl<br/>crazy<br/>cream</div>         | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>credit<br/>creek<br/>crew<br/>cricket<br/>crime<br/>crisp<br/>critic<br/>crop</div>       | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>cross<br/>crouch<br/>crowd<br/>crucial<br/>cruel<br/>cruise<br/>crumble<br/>crunch</div>      | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>crush<br/>cry<br/>crystal<br/>cube<br/>culture<br/>cup<br/>cupboard<br/>curious</div>     | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>current<br/>curtain<br/>curve<br/>cushion<br/>custom<br/>cute<br/>cycle<br/>dad</div>           | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>damage<br/>damp<br/>dance<br/>danger<br/>daring<br/>dash<br/>daughter<br/>dawn</div>           |
| <div>346</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>day<br/>deal<br/>debate<br/>debris<br/>decade<br/>december<br/>decide<br/>decline</div>   | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>decorate<br/>decrease<br/>deer<br/>defense<br/>define<br/>defy<br/>degree<br/>delay</div>     | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>deliver<br/>demand<br/>demise<br/>denial<br/>dentist<br/>deny<br/>depart<br/>depend</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>deposit<br/>depth<br/>deputy<br/>derive<br/>describe<br/>desert<br/>design<br/>desk</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>despair<br/>destroy<br/>detail<br/>detect<br/>develop<br/>device<br/>devote<br/>diagram</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>dial<br/>diamond<br/>diary<br/>dice<br/>diesel<br/>diet<br/>differ<br/>digital</div>      | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>dignity<br/>dilemma<br/>dinner<br/>dinosaur<br/>direct<br/>dirt<br/>disagree<br/>discover</div> | <div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div><div>678</div></div> <div>disease<br/>dish<br/>dismiss<br/>disorder<br/>display<br/>distance<br/>divert<br/>divide</div> |

|                |                                                                                                                                                       |                                                                                                                                                        |                                                                                                                                                         |                                                                                                                                                          |                                                                                                                                                      |                                                                                                                                                       |                                                                                                                                                       |                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <div>678</div>                                                                                                                                        | <div>678</div>                                                                                                                                         | <div>678</div>                                                                                                                                          | <div>678</div>                                                                                                                                           | <div>678</div>                                                                                                                                       | <div>678</div>                                                                                                                                        | <div>678</div>                                                                                                                                        |                                                                                                                                                           |
| <div>345</div> | <div><div>divorce</div><div>dizzy</div><div>doctor</div><div>document</div><div>dog</div><div>doll</div><div>dolphin</div><div>domain</div></div>     | <div><div>donate</div><div>donkey</div><div>donor</div><div>door</div><div>dose</div><div>double</div><div>dove</div><div>draft</div></div>            | <div><div>dragon</div><div>drama</div><div>drastic</div><div>draw</div><div>dream</div><div>dress</div><div>drift</div><div>drill</div></div>           | <div><div>drink</div><div>drip</div><div>drive</div><div>drop</div><div>drum</div><div>dry</div><div>duck</div><div>dumb</div></div>                     | <div><div>dune</div><div>during</div><div>dust</div><div>dutch</div><div>duty</div><div>dwarf</div><div>dynamic</div><div>eager</div></div>          | <div><div>eagle</div><div>early</div><div>earn</div><div>earth</div><div>easily</div><div>east</div><div>easy</div><div>echo</div></div>              | <div><div>ecology</div><div>economy</div><div>edge</div><div>edit</div><div>educate</div><div>effort</div><div>egg</div><div>eight</div></div>        | <div><div>either</div><div>elbow</div><div>elder</div><div>electric</div><div>elegant</div><div>element</div><div>elephant</div><div>elevator</div></div> |
|                | <div><div>elite</div><div>else</div><div>embark</div><div>embody</div><div>embrace</div><div>emerge</div><div>emotion</div><div>employ</div></div>    | <div><div>empower</div><div>empty</div><div>enable</div><div>enact</div><div>end</div><div>endless</div><div>endorse</div><div>enemy</div></div>       | <div><div>energy</div><div>enforce</div><div>engage</div><div>engine</div><div>enhance</div><div>enjoy</div><div>enlist</div><div>enough</div></div>    | <div><div>enrich</div><div>enroll</div><div>ensure</div><div>enter</div><div>entire</div><div>entry</div><div>envelope</div><div>episode</div></div>     | <div><div>equal</div><div>equip</div><div>era</div><div>erase</div><div>erode</div><div>erosion</div><div>error</div><div>erupt</div></div>          | <div><div>escape</div><div>essay</div><div>essence</div><div>estate</div><div>eternal</div><div>ethics</div><div>evidence</div><div>evil</div></div>  | <div><div>evoke</div><div>evolve</div><div>exact</div><div>example</div><div>excess</div><div>exchange</div><div>excite</div><div>exclude</div></div> | <div><div>excuse</div><div>execute</div><div>exercise</div><div>exhaust</div><div>exhibit</div><div>exile</div><div>exist</div><div>exit</div></div>      |
|                | <div><div>exotic</div><div>expand</div><div>expect</div><div>expire</div><div>explain</div><div>expose</div><div>express</div><div>extend</div></div> | <div><div>extra</div><div>eye</div><div>eyebrow</div><div>fabric</div><div>face</div><div>faculty</div><div>fade</div><div>faint</div></div>           | <div><div>faith</div><div>fall</div><div>false</div><div>fame</div><div>family</div><div>famous</div><div>fan</div><div>fancy</div></div>               | <div><div>fantasy</div><div>farm</div><div>fashion</div><div>fat</div><div>fatal</div><div>father</div><div>fatigue</div><div>fault</div></div>          | <div><div>favorite</div><div>feature</div><div>february</div><div>federal</div><div>fee</div><div>feed</div><div>feel</div><div>female</div></div>   | <div><div>fence</div><div>festival</div><div>fetch</div><div>fever</div><div>few</div><div>fiber</div><div>fiction</div><div>field</div></div>        | <div><div>figure</div><div>file</div><div>film</div><div>filter</div><div>final</div><div>find</div><div>fine</div><div>finger</div></div>            | <div><div>finish</div><div>fire</div><div>firm</div><div>first</div><div>fiscal</div><div>fish</div><div>fit</div><div>fitness</div></div>                |
|                | <div><div>fix</div><div>flag</div><div>flame</div><div>flash</div><div>flat</div><div>flavor</div><div>flee</div><div>flight</div></div>              | <div><div>flip</div><div>float</div><div>flock</div><div>floor</div><div>flower</div><div>fluid</div><div>flush</div><div>fly</div></div>              | <div><div>foam</div><div>focus</div><div>fog</div><div>foil</div><div>fold</div><div>follow</div><div>food</div><div>foot</div></div>                   | <div><div>force</div><div>forest</div><div>forget</div><div>fork</div><div>fortune</div><div>forum</div><div>forward</div><div>fossil</div></div>        | <div><div>foster</div><div>found</div><div>fox</div><div>fragile</div><div>frame</div><div>frequent</div><div>fresh</div><div>friend</div></div>     | <div><div>fringe</div><div>frog</div><div>front</div><div>frost</div><div>frown</div><div>frozen</div><div>fruit</div><div>fuel</div></div>           | <div><div>fun</div><div>funny</div><div>furnace</div><div>fury</div><div>future</div><div>gadget</div><div>gain</div><div>galaxy</div></div>          | <div><div>gallery</div><div>game</div><div>gap</div><div>garage</div><div>garbage</div><div>garden</div><div>garlic</div><div>garment</div></div>         |
|                | <div><div>gas</div><div>gasp</div><div>gate</div><div>gather</div><div>gauge</div><div>gaze</div><div>general</div><div>genius</div></div>            | <div><div>genre</div><div>gentle</div><div>genuine</div><div>gesture</div><div>ghost</div><div>giant</div><div>gift</div><div>giggle</div></div>       | <div><div>ginger</div><div>giraffe</div><div>girl</div><div>give</div><div>glad</div><div>glance</div><div>glare</div><div>glass</div></div>            | <div><div>glide</div><div>glimpse</div><div>globe</div><div>gloom</div><div>glory</div><div>glove</div><div>glow</div><div>glue</div></div>              | <div><div>goat</div><div>goddess</div><div>gold</div><div>good</div><div>goose</div><div>gorilla</div><div>gospel</div><div>gossip</div></div>       | <div><div>govern</div><div>gown</div><div>grab</div><div>grace</div><div>grain</div><div>grant</div><div>grape</div><div>grass</div></div>            | <div><div>gravity</div><div>great</div><div>green</div><div>grid</div><div>grief</div><div>grit</div><div>grocery</div><div>group</div></div>         | <div><div>grow</div><div>grunt</div><div>guard</div><div>guess</div><div>guide</div><div>guilt</div><div>guitar</div><div>gun</div></div>                 |
|                | <div><div>gym</div><div>habit</div><div>hair</div><div>half</div><div>hammer</div><div>hamster</div><div>hand</div><div>happy</div></div>             | <div><div>harbor</div><div>hard</div><div>harsh</div><div>harvest</div><div>hat</div><div>have</div><div>hawk</div><div>hazard</div></div>             | <div><div>head</div><div>health</div><div>heart</div><div>heavy</div><div>hedghog</div><div>height</div><div>hello</div><div>helmet</div></div>         | <div><div>help</div><div>hen</div><div>hero</div><div>hidden</div><div>high</div><div>hill</div><div>hint</div><div>hip</div></div>                      | <div><div>hire</div><div>history</div><div>hobby</div><div>hockey</div><div>hold</div><div>hole</div><div>holiday</div><div>hollow</div></div>       | <div><div>home</div><div>honey</div><div>hood</div><div>hope</div><div>horn</div><div>horror</div><div>horse</div><div>hospital</div></div>           | <div><div>host</div><div>hotel</div><div>hour</div><div>hover</div><div>hub</div><div>huge</div><div>human</div><div>humble</div></div>               | <div><div>humor</div><div>hundred</div><div>hungry</div><div>hunt</div><div>hurdle</div><div>hurry</div><div>hurt</div><div>husband</div></div>           |
|                | <div><div>hybrid</div><div>ice</div><div>icon</div><div>idea</div><div>identify</div><div>idle</div><div>ignore</div><div>ill</div></div>             | <div><div>illegal</div><div>illness</div><div>image</div><div>imitate</div><div>immense</div><div>immune</div><div>impact</div><div>impose</div></div> | <div><div>improve</div><div>impulse</div><div>inch</div><div>include</div><div>income</div><div>increase</div><div>index</div><div>indicate</div></div> | <div><div>indoor</div><div>industry</div><div>infant</div><div>inflict</div><div>inform</div><div>inhale</div><div>inherit</div><div>initial</div></div> | <div><div>inject</div><div>injury</div><div>inmate</div><div>inner</div><div>innocent</div><div>input</div><div>inquiry</div><div>insane</div></div> | <div><div>insect</div><div>inside</div><div>inspire</div><div>install</div><div>intact</div><div>interest</div><div>into</div><div>invest</div></div> | <div><div>invite</div><div>involve</div><div>iron</div><div>island</div><div>isolate</div><div>issue</div><div>item</div><div>ivory</div></div>       | <div><div>jacket</div><div>jaguar</div><div>jar</div><div>jazz</div><div>jealous</div><div>jeans</div><div>jelly</div><div>jewel</div></div>              |
|                | <div><div>job</div><div>join</div><div>joke</div><div>journey</div><div>joy</div><div>judge</div><div>juice</div><div>jump</div></div>                | <div><div>jungle</div><div>junior</div><div>junk</div><div>just</div><div>kangaroo</div><div>keen</div><div>keep</div><div>ketchup</div></div>         | <div><div>key</div><div>kick</div><div>kid</div><div>kidney</div><div>kind</div><div>kingdom</div><div>kiss</div><div>kit</div></div>                   | <div><div>kitchen</div><div>kite</div><div>kitten</div><div>kiwi</div><div>knee</div><div>knife</div><div>knock</div><div>know</div></div>               | <div><div>lab</div><div>label</div><div>labor</div><div>ladder</div><div>lady</div><div>lake</div><div>lamp</div><div>language</div></div>           | <div><div>laptop</div><div>large</div><div>later</div><div>latin</div><div>laugh</div><div>laundry</div><div>lava</div><div>law</div></div>           | <div><div>lawn</div><div>lawsuit</div><div>layer</div><div>lazy</div><div>leader</div><div>leaf</div><div>learn</div><div>leave</div></div>           | <div><div>lecture</div><div>left</div><div>leg</div><div>legal</div><div>legend</div><div>leisure</div><div>lemon</div><div>lend</div></div>              |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div><div>length</div><div>lens</div><div>leopard</div><div>lesson</div><div>letter</div><div>level</div><div>liar</div><div>liberty</div></div></div>       | <div><div><div>library</div><div>license</div><div>life</div><div>lift</div><div>light</div><div>like</div><div>limb</div><div>limit</div></div></div>             | <div><div><div>link</div><div>lion</div><div>liquid</div><div>list</div><div>little</div><div>live</div><div>lizard</div><div>load</div></div></div>             | <div><div><div>loan</div><div>lobster</div><div>local</div><div>lock</div><div>logic</div><div>lonely</div><div>long</div><div>loop</div></div></div>              | <div><div><div>lottery</div><div>loud</div><div>lounge</div><div>love</div><div>loyal</div><div>lucky</div><div>luggage</div><div>lumber</div></div></div>          | <div><div><div>lunar</div><div>lunch</div><div>luxury</div><div>lyrics</div><div>machine</div><div>mad</div><div>magic</div><div>magnet</div></div></div>      | <div><div><div>maid</div><div>mail</div><div>main</div><div>major</div><div>make</div><div>mammal</div><div>man</div><div>manage</div></div></div>           | <div><div><div>mandate</div><div>mango</div><div>mansion</div><div>manual</div><div>maple</div><div>marble</div><div>march</div><div>margin</div></div></div>         |
| <div><div><div>marine</div><div>market</div><div>marriage</div><div>mask</div><div>mass</div><div>master</div><div>match</div><div>material</div></div></div>     | <div><div><div>math</div><div>matrix</div><div>matter</div><div>maximum</div><div>maze</div><div>meadow</div><div>mean</div><div>measure</div></div></div>         | <div><div><div>meat</div><div>mechanic</div><div>medal</div><div>media</div><div>melody</div><div>melt</div><div>member</div><div>memory</div></div></div>       | <div><div><div>mention</div><div>menu</div><div>mercy</div><div>merge</div><div>merit</div><div>merry</div><div>mesh</div><div>message</div></div></div>           | <div><div><div>metal</div><div>method</div><div>middle</div><div>midnight</div><div>milk</div><div>million</div><div>mimic</div><div>mind</div></div></div>         | <div><div><div>minimum</div><div>minor</div><div>minute</div><div>miracle</div><div>mirror</div><div>misery</div><div>miss</div><div>mistake</div></div></div> | <div><div><div>mix</div><div>mixed</div><div>mixture</div><div>mobile</div><div>model</div><div>modify</div><div>mom</div><div>moment</div></div></div>      | <div><div><div>monitor</div><div>monkey</div><div>monster</div><div>month</div><div>moon</div><div>moral</div><div>more</div><div>morning</div></div></div>           |
| <div><div><div>mosquito</div><div>mother</div><div>motion</div><div>motor</div><div>mountain</div><div>mouse</div><div>move</div><div>movie</div></div></div>     | <div><div><div>much</div><div>muffin</div><div>mule</div><div>multiply</div><div>muscle</div><div>museum</div><div>mushroom</div><div>music</div></div></div>      | <div><div><div>must</div><div>mutual</div><div>myself</div><div>mystery</div><div>myth</div><div>naive</div><div>name</div><div>napkin</div></div></div>         | <div><div><div>narrow</div><div>nasty</div><div>nation</div><div>nature</div><div>near</div><div>neck</div><div>need</div><div>negative</div></div></div>          | <div><div><div>neglect</div><div>neither</div><div>nephew</div><div>nerve</div><div>nest</div><div>net</div><div>network</div><div>neutral</div></div></div>        | <div><div><div>never</div><div>news</div><div>next</div><div>nice</div><div>night</div><div>noble</div><div>noise</div><div>nominee</div></div></div>          | <div><div><div>noodle</div><div>normal</div><div>north</div><div>nose</div><div>notable</div><div>note</div><div>nothing</div><div>notice</div></div></div>  | <div><div><div>novel</div><div>now</div><div>nuclear</div><div>number</div><div>nurse</div><div>nut</div><div>oak</div><div>obey</div></div></div>                    |
| <div><div><div>object</div><div>oblige</div><div>obscure</div><div>observe</div><div>obtain</div><div>obvious</div><div>occur</div><div>ocean</div></div></div>   | <div><div><div>october</div><div>odor</div><div>off</div><div>offer</div><div>office</div><div>often</div><div>oil</div><div>okay</div></div></div>                | <div><div><div>old</div><div>olive</div><div>olympic</div><div>omit</div><div>once</div><div>one</div><div>onion</div><div>online</div></div></div>              | <div><div><div>only</div><div>open</div><div>opera</div><div>opinion</div><div>oppose</div><div>option</div><div>orange</div><div>orbit</div></div></div>          | <div><div><div>orchard</div><div>order</div><div>ordinary</div><div>organ</div><div>orient</div><div>original</div><div>orphan</div><div>ostrich</div></div></div>  | <div><div><div>other</div><div>outdoor</div><div>outer</div><div>output</div><div>outside</div><div>oval</div><div>oven</div><div>over</div></div></div>       | <div><div><div>own</div><div>owner</div><div>oxygen</div><div>oyster</div><div>ozone</div><div>pact</div><div>paddle</div><div>page</div></div></div>        | <div><div><div>pair</div><div>palace</div><div>palm</div><div>panda</div><div>panel</div><div>panic</div><div>panther</div><div>paper</div></div></div>               |
| <div><div><div>parade</div><div>parent</div><div>park</div><div>parrot</div><div>party</div><div>pass</div><div>patch</div><div>path</div></div></div>            | <div><div><div>patient</div><div>patrol</div><div>pattern</div><div>pause</div><div>pave</div><div>payment</div><div>peace</div><div>peanut</div></div></div>      | <div><div><div>pear</div><div>peasant</div><div>pelican</div><div>pen</div><div>penalty</div><div>pencil</div><div>people</div><div>pepper</div></div></div>     | <div><div><div>perfect</div><div>permit</div><div>person</div><div>pet</div><div>phone</div><div>photo</div><div>phrase</div><div>physical</div></div></div>       | <div><div><div>piano</div><div>picnic</div><div>picture</div><div>piece</div><div>pig</div><div>pigeon</div><div>pill</div><div>pilot</div></div></div>             | <div><div><div>pink</div><div>pioneer</div><div>pipe</div><div>pistol</div><div>pitch</div><div>pizza</div><div>place</div><div>planet</div></div></div>       | <div><div><div>plastic</div><div>plate</div><div>play</div><div>please</div><div>pledge</div><div>pluck</div><div>plug</div><div>plunge</div></div></div>    | <div><div><div>poem</div><div>poet</div><div>point</div><div>polar</div><div>pole</div><div>police</div><div>pond</div><div>pony</div></div></div>                    |
| <div><div><div>pool</div><div>popular</div><div>portion</div><div>position</div><div>possible</div><div>post</div><div>potato</div><div>pottery</div></div></div> | <div><div><div>poverty</div><div>powder</div><div>power</div><div>practice</div><div>praise</div><div>predict</div><div>prefer</div><div>prepare</div></div></div> | <div><div><div>present</div><div>pretty</div><div>prevent</div><div>price</div><div>pride</div><div>primary</div><div>print</div><div>priority</div></div></div> | <div><div><div>prison</div><div>private</div><div>prize</div><div>problem</div><div>process</div><div>produce</div><div>profit</div><div>program</div></div></div> | <div><div><div>project</div><div>promote</div><div>proof</div><div>property</div><div>prosper</div><div>protect</div><div>proud</div><div>provide</div></div></div> | <div><div><div>public</div><div>pudding</div><div>pull</div><div>pulp</div><div>pulse</div><div>pumpkin</div><div>punch</div><div>pupil</div></div></div>      | <div><div><div>puppy</div><div>purchase</div><div>purity</div><div>purpose</div><div>purse</div><div>push</div><div>put</div><div>puzzle</div></div></div>   | <div><div><div>pyramid</div><div>quality</div><div>quantum</div><div>quarter</div><div>question</div><div>quick</div><div>quit</div><div>quiz</div></div></div>       |
| <div><div><div>quote</div><div>rabbit</div><div>raccoon</div><div>race</div><div>rack</div><div>radar</div><div>radio</div><div>rail</div></div></div>            | <div><div><div>rain</div><div>raise</div><div>rally</div><div>ramp</div><div>ranch</div><div>random</div><div>range</div><div>rapid</div></div></div>              | <div><div><div>rare</div><div>rate</div><div>rather</div><div>raven</div><div>raw</div><div>razor</div><div>ready</div><div>real</div></div></div>               | <div><div><div>reason</div><div>rebel</div><div>rebuild</div><div>recall</div><div>receive</div><div>recipe</div><div>record</div><div>recycle</div></div></div>   | <div><div><div>reduce</div><div>reflect</div><div>reform</div><div>refuse</div><div>region</div><div>regret</div><div>regular</div><div>reject</div></div></div>    | <div><div><div>relax</div><div>release</div><div>relief</div><div>rely</div><div>remain</div><div>remember</div><div>remind</div><div>remove</div></div></div> | <div><div><div>render</div><div>renew</div><div>rent</div><div>reopen</div><div>repair</div><div>repeat</div><div>replace</div><div>report</div></div></div> | <div><div><div>require</div><div>rescue</div><div>resemble</div><div>resist</div><div>resource</div><div>response</div><div>result</div><div>retire</div></div></div> |
| <div><div><div>retreat</div><div>return</div><div>reunion</div><div>reveal</div><div>review</div><div>reward</div><div>rhythm</div><div>rib</div></div></div>     | <div><div><div>ribbon</div><div>rice</div><div>rich</div><div>ride</div><div>ridge</div><div>rifle</div><div>right</div><div>rigid</div></div></div>               | <div><div><div>ring</div><div>riot</div><div>ripple</div><div>risk</div><div>ritual</div><div>rival</div><div>river</div><div>road</div></div></div>             | <div><div><div>roast</div><div>robot</div><div>robust</div><div>rocket</div><div>romance</div><div>roof</div><div>rookie</div><div>room</div></div></div>          | <div><div><div>rose</div><div>rotate</div><div>rough</div><div>round</div><div>route</div><div>royal</div><div>rubber</div><div>rude</div></div></div>              | <div><div><div>rug</div><div>rule</div><div>run</div><div>runway</div><div>rural</div><div>sad</div><div>saddle</div><div>sadness</div></div></div>            | <div><div><div>safe</div><div>sail</div><div>salad</div><div>salmon</div><div>salon</div><div>salt</div><div>salute</div><div>same</div></div></div>         | <div><div><div>sample</div><div>sand</div><div>satisfy</div><div>satoshi</div><div>sauce</div><div>sausage</div><div>save</div><div>say</div></div></div>             |

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| <div><div>scale</div><div>scan</div><div>scare</div><div>scatter</div><div>scene</div><div>scheme</div><div>school</div><div>science</div></div>   | <div><div>scissors</div><div>scorpion</div><div>scout</div><div>scrap</div><div>screen</div><div>script</div><div>scrub</div><div>sea</div></div> | <div><div>search</div><div>season</div><div>seat</div><div>second</div><div>secret</div><div>section</div><div>security</div><div>seed</div></div>      | <div><div>seek</div><div>segment</div><div>select</div><div>sell</div><div>seminar</div><div>senior</div><div>sense</div><div>sentence</div></div>   | <div><div>series</div><div>service</div><div>session</div><div>settle</div><div>setup</div><div>seven</div><div>shadow</div><div>shaft</div></div> | <div><div>shallow</div><div>share</div><div>shed</div><div>shell</div><div>sheriff</div><div>shield</div><div>shift</div><div>shine</div></div>        | <div><div>ship</div><div>shiver</div><div>shock</div><div>shoe</div><div>shoot</div><div>shop</div><div>short</div><div>shoulder</div></div>       | <div><div>shove</div><div>shrimp</div><div>shrug</div><div>shuffle</div><div>shy</div><div>sibling</div><div>sick</div><div>side</div></div>           |
| <div><div>siege</div><div>sight</div><div>sign</div><div>silent</div><div>silk</div><div>silly</div><div>silver</div><div>similar</div></div>      | <div><div>simple</div><div>since</div><div>sing</div><div>siren</div><div>sister</div><div>situate</div><div>six</div><div>size</div></div>       | <div><div>skate</div><div>sketch</div><div>ski</div><div>skill</div><div>skin</div><div>skirt</div><div>skull</div><div>slab</div></div>                | <div><div>slam</div><div>sleep</div><div>slender</div><div>slice</div><div>slide</div><div>slight</div><div>slim</div><div>slogan</div></div>        | <div><div>slot</div><div>slow</div><div>slush</div><div>small</div><div>smart</div><div>smile</div><div>smoke</div><div>smooth</div></div>         | <div><div>snack</div><div>snake</div><div>snap</div><div>sniff</div><div>snow</div><div>soap</div><div>soccer</div><div>social</div></div>             | <div><div>sock</div><div>soda</div><div>soft</div><div>solar</div><div>soldier</div><div>solid</div><div>solution</div><div>solve</div></div>      | <div><div>someone</div><div>song</div><div>soon</div><div>sorry</div><div>sort</div><div>soul</div><div>sound</div><div>soup</div></div>               |
| <div><div>source</div><div>south</div><div>space</div><div>spare</div><div>spatial</div><div>spawn</div><div>speak</div><div>special</div></div>   | <div><div>speed</div><div>spell</div><div>spend</div><div>sphere</div><div>spice</div><div>spider</div><div>spike</div><div>spin</div></div>      | <div><div>spirit</div><div>split</div><div>spoil</div><div>sponsor</div><div>spoon</div><div>sport</div><div>spot</div><div>spray</div></div>           | <div><div>spread</div><div>spring</div><div>spy</div><div>square</div><div>squeeze</div><div>squirrel</div><div>stable</div><div>stadium</div></div> | <div><div>staff</div><div>stage</div><div>stairs</div><div>stamp</div><div>stand</div><div>start</div><div>state</div><div>stay</div></div>        | <div><div>steak</div><div>steel</div><div>stem</div><div>step</div><div>stereo</div><div>stick</div><div>still</div><div>sting</div></div>             | <div><div>stock</div><div>stomach</div><div>stone</div><div>stool</div><div>story</div><div>stove</div><div>strategy</div><div>street</div></div>  | <div><div>strike</div><div>strong</div><div>struggle</div><div>student</div><div>stuff</div><div>stumble</div><div>style</div><div>subject</div></div> |
| <div><div>submit</div><div>subway</div><div>success</div><div>such</div><div>sudden</div><div>suffer</div><div>sugar</div><div>suggest</div></div> | <div><div>suit</div><div>summer</div><div>sun</div><div>sunny</div><div>sunset</div><div>super</div><div>supply</div><div>supreme</div></div>     | <div><div>sure</div><div>surface</div><div>surge</div><div>surprise</div><div>surround</div><div>survey</div><div>suspect</div><div>sustain</div></div> | <div><div>swallow</div><div>swamp</div><div>swap</div><div>swarm</div><div>swear</div><div>sweet</div><div>swift</div><div>swim</div></div>          | <div><div>swing</div><div>switch</div><div>sword</div><div>symbol</div><div>symptom</div><div>syrup</div><div>system</div><div>table</div></div>   | <div><div>tackle</div><div>tag</div><div>tail</div><div>talent</div><div>talk</div><div>tank</div><div>tape</div><div>target</div></div>               | <div><div>task</div><div>taste</div><div>tattoo</div><div>taxi</div><div>teach</div><div>team</div><div>tell</div><div>ten</div></div>             | <div><div>tenant</div><div>tennis</div><div>tent</div><div>term</div><div>test</div><div>text</div><div>thank</div><div>that</div></div>               |
| <div><div>theme</div><div>then</div><div>theory</div><div>there</div><div>they</div><div>thing</div><div>this</div><div>thought</div></div>        | <div><div>three</div><div>thrive</div><div>throw</div><div>thumb</div><div>thunder</div><div>ticket</div><div>tide</div><div>tiger</div></div>    | <div><div>tilt</div><div>timber</div><div>time</div><div>tiny</div><div>tip</div><div>tired</div><div>tissue</div><div>title</div></div>                | <div><div>toast</div><div>tobacco</div><div>today</div><div>toddler</div><div>toe</div><div>together</div><div>toilet</div><div>token</div></div>    | <div><div>tomato</div><div>tomorrow</div><div>tone</div><div>tongue</div><div>tonight</div><div>tool</div><div>tooth</div><div>top</div></div>     | <div><div>topic</div><div>topple</div><div>torch</div><div>tornado</div><div>tortoise</div><div>toss</div><div>total</div><div>tourist</div></div>     | <div><div>toward</div><div>tower</div><div>town</div><div>toy</div><div>track</div><div>trade</div><div>traffic</div><div>tragic</div></div>       | <div><div>train</div><div>transfer</div><div>trap</div><div>trash</div><div>travel</div><div>tray</div><div>treat</div><div>tree</div></div>           |
| <div><div>trend</div><div>trial</div><div>tribe</div><div>trick</div><div>trigger</div><div>trim</div><div>trip</div><div>trophy</div></div>       | <div><div>trouble</div><div>truck</div><div>true</div><div>truly</div><div>trumpet</div><div>trust</div><div>truth</div><div>try</div></div>      | <div><div>tube</div><div>tuition</div><div>tumble</div><div>tuna</div><div>tunnel</div><div>turkey</div><div>turn</div><div>turtle</div></div>          | <div><div>twelve</div><div>twenty</div><div>twice</div><div>twin</div><div>twist</div><div>two</div><div>type</div><div>typical</div></div>          | <div><div>ugly</div><div>umbrella</div><div>unable</div><div>unaware</div><div>uncle</div><div>uncover</div><div>under</div><div>undo</div></div>  | <div><div>unfair</div><div>unfold</div><div>unhappy</div><div>uniform</div><div>unique</div><div>unit</div><div>universe</div><div>unknown</div></div> | <div><div>unlock</div><div>until</div><div>unusual</div><div>unveil</div><div>update</div><div>upgrade</div><div>uphold</div><div>upon</div></div> | <div><div>upper</div><div>upset</div><div>urban</div><div>urge</div><div>usage</div><div>use</div><div>used</div><div>useful</div></div>               |
| <div><div>useless</div><div>usual</div><div>utility</div><div>vacant</div><div>vacuum</div><div>vague</div><div>valid</div><div>valley</div></div> | <div><div>valve</div><div>van</div><div>vanish</div><div>vapor</div><div>various</div><div>vast</div><div>vault</div><div>vehicle</div></div>     | <div><div>velvet</div><div>vendor</div><div>venture</div><div>venue</div><div>verb</div><div>verify</div><div>version</div><div>very</div></div>        | <div><div>vessel</div><div>veteran</div><div>viable</div><div>vibrant</div><div>vicious</div><div>victory</div><div>video</div><div>view</div></div> | <div><div>village</div><div>vintage</div><div>violin</div><div>virtual</div><div>virus</div><div>visa</div><div>visit</div><div>visual</div></div> | <div><div>vital</div><div>vivid</div><div>vocal</div><div>voice</div><div>void</div><div>volcano</div><div>volume</div><div>vote</div></div>           | <div><div>voyage</div><div>wage</div><div>wagon</div><div>wait</div><div>walk</div><div>wall</div><div>walnut</div><div>want</div></div>           | <div><div>warfare</div><div>warm</div><div>warrior</div><div>wash</div><div>wasp</div><div>waste</div><div>water</div><div>wave</div></div>            |
| <div><div>way</div><div>wealth</div><div>weapon</div><div>wear</div><div>weasel</div><div>weather</div><div>web</div><div>wedding</div></div>      | <div><div>weekend</div><div>weird</div><div>welcome</div><div>west</div><div>wet</div><div>whale</div><div>what</div><div>wheat</div></div>       | <div><div>wheel</div><div>when</div><div>where</div><div>whip</div><div>whisper</div><div>wide</div><div>width</div><div>wife</div></div>               | <div><div>wild</div><div>will</div><div>win</div><div>window</div><div>wine</div><div>wing</div><div>wink</div><div>winner</div></div>               | <div><div>winter</div><div>wire</div><div>wisdom</div><div>wise</div><div>wish</div><div>witness</div><div>wolf</div><div>woman</div></div>        | <div><div>wonder</div><div>wood</div><div>wool</div><div>word</div><div>work</div><div>world</div><div>worry</div><div>worth</div></div>               | <div><div>wrap</div><div>wreck</div><div>wrestle</div><div>wrist</div><div>write</div><div>wrong</div><div>yard</div><div>year</div></div>         | <div><div>yellow</div><div>you</div><div>young</div><div>youth</div><div>zebra</div><div>zero</div><div>zone</div><div>zoo</div></div>                 |

# BIP39 Printable Wordlist — Manual

## Why

Software-based seed generation has a large attack surface: compromised RNGs, malware, clipboard sniffers, screen capture, or a backdoored hardware stack. Even an air-gapped computer still requires trusting its entire software and hardware chain.

Physical coin flips are a verifiable entropy source — the randomness is directly observable. This also avoids having to trust the random-number generator inside a hardware wallet or other device. If that RNG is flawed or badly implemented, the generated seed may be weaker than it appears. With this printable layout, you can go from coin flips to words on paper, with no screen or device involved in the generation step.

## How to use the coin flips

Flip 11 coins. Place them left to right into the slots below. Then match each group against the dark/light blocks on the lookup pages. You do not need to convert anything by hand — just match the patterns.

Do not reroll just because the result looks too uneven, too balanced, or otherwise unusual. A real random process naturally produces streaks and lopsided results. Rejecting those outcomes would bias the space of possible seeds.

What matters is that the flipping and placement process itself is neutral. A good method: shake all 11 coins in a closed box, spill them onto the table without looking, then slowly sweep a ruler across the spread in a fixed direction (e.g. top to bottom). The order in which the ruler's edge first touches each coin defines the bit order — first touched = bit 1, second = bit 2, and so on through bit 11. Only then read off heads/tails in that fixed order. Do not spin coins on a table — spun coins are biased by their center of mass and can land one way 70%+ of the time. Do not subconsciously pick coins based on how they landed, and do not reorder them after seeing the faces.

| page | bits 1–2                                        | column                                            | bits 6–8  |
|------|-------------------------------------------------|---------------------------------------------------|-----------|
|      | <div><div>1</div><div>2</div></div>             | <div><div>6</div><div>7</div><div>8</div></div>   |           |
| row  | bits 3–5                                        | word                                              | bits 9–11 |
|      | <div><div>3</div><div>4</div><div>5</div></div> | <div><div>9</div><div>10</div><div>11</div></div> |           |

●

 heads = dark = 1

●

 tails = light = 0

## How to create a wallet

You can build a BIP39 wallet entirely from your own coin flips, instead of trusting the random-number generator inside a hardware wallet. The idea is simple: use the lookup pages to turn each batch of 11 coin flips into one BIP39 word, repeat for every word in the phrase, and then have a hardware wallet import that phrase as if it had generated it itself.

There is one subtlety. BIP39 does not let every word be fully random. A 12-word phrase encodes 128 bits of your own entropy plus a 4-bit checksum that BIP39 computes from those 128 bits. The first 11 words are  $11 \times 11 = 121$  bits — fully yours. The 12th word holds the remaining 7 bits of entropy plus the 4 checksum bits, all packed into its 11 bits. So the 12th word is partly random (your last 7 coin flips) and partly determined (the 4 checksum bits, which you cannot choose).

A practical workflow:

- Flip 11 coins for each of the first 11 words and look them up on the sheet.
- Flip 11 coins for the 12th word as well, but treat the result as a starting point. Look up the word the sheet gives you.
- Keep its first 7 bits fixed — the last 4 are the BIP39 checksum.
- On a trusted hardware wallet (e.g. a Trezor), start the recovery / import flow and enter your phrase. Try each of the 16 candidates as the 12th word until the device accepts the phrase. A wrong checksum is rejected immediately; the accepted phrase is your real mnemonic.

The wallet is not adding randomness — your flips already fixed the entropy; it only checks the checksum and derives the wallet from your phrase. The seed becomes digital at this step, which is unavoidable: the goal here is to keep the generation off any device, not the seed's whole lifetime. For the same reason, do the checksum check via the wallet's own restore flow, not a website or other software (unless fully offline and trusted).

Longer phrases work the same way, with more checksum bits in the last word: 15 words → 32 candidates, 18 → 64, 21 → 128, 24 → 256.

## Backing up the words

This sheet only helps you generate the words. Storing them safely — paper vs. metal, memorization, geographic splits, Shamir secret sharing, inheritance planning — is a separate problem beyond the scope of this project. A seed phrase is permanent: lose it and the funds are unrecoverable; share it (with passphrase, if any) and someone else can spend them.

### Additional detail: exact mapping

You do not need to understand this to use the sheet. This is only for someone who wants to know how the printed layout maps to the exact BIP39 index.

Read the numbered bits from left to right. Heads/dark means 1. Tails/light means 0. Split the 11 bits as 2 | 3 | 3 | 3. Those four groups mean page | row | column | word.

If you want the exact BIP39 list position, read the full 11-bit pattern as a binary number. The BIP39 wordlist counts from 0, so the Nth word uses binary(N-1).

### Example: 1125th word

The 1125th BIP39 word is 'milk'. Take  $1125 - 1 = 1124$ . In 11-bit binary that is 10001100100, or split up: 10 | 001 | 100 | 100.



So on the printed sheet: go to page 10, then row 001, then column 100, then word 100. That lands on 'milk'.